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Evaluation of Derma Genius® Resistance Real-Time Polymerase Chain Reaction (RT-PCR) for Rapid Detection of Terbinafine Resistant *Trichophyton* species

Running Title: DermaGenius® qPCR for *Trichophyton* species.

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Author Contributions

A.C. and G.D. designed the study; A.S., and P.S., performed the experimental work and A.C., A.S., and P.S. analysed the data and drafted the manuscript, A.C., J.F.M., and G.D. edited the manuscript. All authors have read and approved the final manuscript.

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Conflicts of Interest Statement

J.F.M. received grants from Astellas, Basilea, and Merck. He has been a consultant to Astellas, Basilea, Scynexis and Merck and has received speaker's fees from Merck, United Medical, TEVA

and Gilead. G.D. is an employee of Pathonostics, Maastricht, The Netherlands. All other authors declare no potential conflicts of interest. We alone are responsible for the content and writing of the paper.

ABSTRACT

Background: Treatment-resistant dermatophytosis caused by *Trichophyton mentagrophytes/interdigitale* complex has emerged as a global public health threat, particularly in endemic countries like India and has spread to many other countries. This veritable spread is alarming due to increase in resistance to terbinafine, which targets the ergosterol biosynthetic pathway by inhibiting the enzyme squalene epoxidase (SQLE). About two third of studies worldwide have reported amino acid substitutions Phe397Leu and Leu393Phe in the SQLE protein to be responsible for high terbinafine MICs.

Objectives: We evaluated the efficacy of the newly developed DermaGenius® Resistance real-time PCR assay to rapidly identify *Trichophyton* isolates harbouring most common *SQLE* mutant (Phe397Leu and Leu393Phe) conferring high terbinafine resistance from wild-type susceptible isolates.

Methods: A total of 97 *Trichophyton* isolates confirmed by ITS sequencing as *T. mentagrophytes/interdigitale* (recently named *T. indotineae* n=90), *T. rubrum*/*T. soudanense* (n=3), *T. mentagrophytes* (n=2), and *T. tonsurans* (n=2) were analysed to evaluate DermaGenius® Resistance real-time PCR assay. All 40 *T. indotineae* isolates exhibiting amino acid substitutions Phe397Leu or Leu393Phe identified by *SQLE* gene sequencing were evaluated for detection of non-wild type strains by real-time PCR. Antifungal susceptibility testing for terbinafine was done by CLSI microbroth dilution method.

Results: All terbinafine resistant isolates harbouring amino acid substitutions Phe397Leu or Leu393Phe in *SQLE* genes were correctly recorded as *SQLE* mutants by the DermaGenius® Resistance real-time PCR assay.

Conclusions: The DermaGenius® Resistance real-time PCR assay effectively identified *Trichophyton* species and distinguished wild-type from *SQLE* mutant genotype that harbour Phe397Leu and Leu393Phe amino acid substitutions.

Keywords: *Trichophyton mentagrophytes/interdigitale*, *Trichophyton indotineae*, *SQLE*, terbinafine resistance, DermaGenius®, India

Main Text

1. INTRODUCTION

Dermatophytosis is superficial fungal infections of hair, nail and skin caused by three main classes of dermatophytes namely *Epidermophyton*, *Trichophyton*, and *Microsporum*. *Trichophyton* is the most common dermatophyte genus responsible for tinea corporis, affecting glabrous skin.¹ Tinea corporis is commonly treated by the ergosterol biosynthesis inhibitor terbinafine, which inhibits fungal growth by blocking the activity of squalene epoxidase (SQLE) resulting in the accumulation of squalene and depletion of ergosterol from the fungal wall. Over the last decade, an unrelenting threat is posed by ongoing outbreaks of *Trichophyton* infections in Southeast Asia that are resistant to the first line antifungal terbinafine.²⁻⁵ Treatment-resistant dermatophytosis caused by *Trichophyton mentagrophytes/interdigitale* complex has emerged as a global public health threat, particularly in endemic countries like India³⁻⁵ where this disease has spread as an epidemic of difficult-to-treat recalcitrant dermatophytosis. In the last five years terbinafine resistance in dermatophytosis has spread to Iran⁶⁻⁸, Japan⁹, Denmark^{10,11}, Belgium¹², Finland¹³, Switzerland^{14,15}, Germany^{16,17}, USA¹⁸, Canada¹⁹, Bahrain¹⁶, and Brazil²⁰. The rapid global spread of terbinafine resistant dermatophytes is alarming considering that alternative therapeutic options to treat drug resistant dermatophytosis are limited.²¹

The highly terbinafine resistant Indian *Trichophyton* strains have recently been proposed to be a new species i.e. *T. indotineae* independent of *T. mentagrophytes/interdigitale* complex according to clinical and mycological features, multi gene phylogeny, and genome sequencing.^{4,22} Interestingly, the globally encountered most common mechanism of clinical terbinafine resistance in *Trichophyton* species has been associated with mutations in the squalene epoxidase gene.^{14,23,24} In particular, alterations in the specific amino acid hot spots Leu393, Phe397, Phe415, and His440 of squalene epoxidase increase terbinafine MICs of *T. interdigitale/mentagrophytes*.¹⁴ So far, in the 18 clinical studies worldwide, the most dominant substitutions associated with the *SQLE* gene of *Trichophyton* strains^{3-6,8,12-14,16,17,22,25-31} obtained from patients with superficial dermatophytosis found the Phe397Leu amino acid substitution in 15 studies followed by Leu393Phe in 10 studies suggesting that both these variants are dominant mutations causing high rates of terbinafine resistance. Interestingly, both of these substitutions lead to high MIC values, while other reported variants have been associated with lower MIC elevations but still potentially pose a risk of treatment failure.^{14, 26}

Although rising number of *Trichophyton* infections with clinical resistance have led to increased awareness of this emerging problem, a major challenge that still remains in the diagnostic mycology laboratory is that susceptibility testing of dermatophytes is not routinely performed, and thus, resistance might remain unrecognised. Furthermore, broth microdilution susceptibility testing methods require extended incubation periods for dermatophytes, posing a risk of bacterial or mould contamination, which may hamper the correct interpretation of terbinafine MICs. Also, clinical breakpoints (CBP) for different antifungals against dermatophytes have not been defined due to lack of clinical correlation and pharmacokinetic/pharmacodynamic studies. It is important to emphasise that with the increasing burden of terbinafine resistance among *Trichophyton* species, it is necessary to identify resistant isolates early for appropriate management. In the present study we evaluate the newly developed DermaGenius® Resistance real-time PCR assay as a rapid *in vitro* diagnostic method that identifies *Trichophyton* species and detects strains harbouring the two most dominant SQLE amino acid substitutions (Phe397Leu, Leu393Phe) from culture isolates. Further, we analysed the results of species identification of *Trichophyton* detected by DermaGenius® Resistance real-time PCR assay by employing ITS sequencing. The SQLE mutants identified by DermaGenius® Resistance real-time PCR assay were correlated using CLSI broth microdilution antifungal susceptibility testing of terbinafine and *SQLE* gene sequencing and mutation analysis.

2. MATERIALS AND METHODS

A total of 97 *Trichophyton* isolates obtained from recalcitrant dermatophytosis cases of tinea corporis/cruris (n = 86), tinea capitis (n = 10) and onychomycosis (n = 1) were included to evaluate the newly developed DermaGenius® Resistance real-time PCR assay. The isolates were confirmed as *T. indotineae* (n=90), *T. rubrum*/*T. soudanense* (n=3), *T. mentagrophytes* (n=2) and *T. tonsurans* (n=2) by ITS rDNA region gene sequencing. The isolates were subjected to antifungal susceptibility testing using CLSI M38-Ed3³² broth microdilution methodology followed by *SQLE* gene sequencing and mutation analysis.^{3, 4}

The new assay uses melting curve analysis to detect *Trichophyton* species and differentiate between ‘wildtype’ and the most clinically relevant SQLE mutant genotypes, including two most dominant Phe397Leu, and Leu393Phe amino acid substitutions along with less prevalent Leu393Ser, Phe397Ile, Phe397Val amino acid substitutions.

2.1 DNA extraction: DNA extraction was performed using the NucliSENS easyMAG (bioMérieux, Amersfoort, the Netherlands) as per manufacturer's instructions. Briefly, small pieces of mycelium were collected with a sterile scalpel from the SDA plates and transferred to green bead tubes (Roche Diagnostics, Indianapolis, IN, USA) and 500 µl of NucliSENS lysis buffer (bioMérieux, Amersfoort, the Netherlands). The mycelium suspension was subsequently bead-beaten in a MagNALyser instrument (Roche Diagnostics) for 45 s at 6,500 rpm. Proteinase K (Roche Diagnostics) was added, incubated for 10 min at 65 °C, and subsequently inactivated for 10 min at 95 °C. After centrifugation, the supernatant of the lysed mycelium suspension was transferred to a new tube together with 5 µl of internal control and DNA was extracted using the NucliSENS easyMAG Extraction system (bioMérieux). DNA was stored at -20 °C until testing with the DermaGenius® Resistance real-time PCR kit which is designed to detect SQLE mutants (Phe397Leu, Leu393Phe, Leu393Ser, Phe397Ile, and Phe397Val) and thus differentiate terbinafine resistant from wild type (susceptible) isolates. In addition, target specific probes were included in the multiplex PCR to differentiate most clinically relevant *Trichophyton* species as detailed below.

2.2 Species level identification: For the real-time PCR, DNA samples were processed according to manufacturer's instructions: 5 µl of DNA extract was added to the PCR mix, and a Mic qPCR cyclor (Bio Molecular Systems, Australia) was used for amplification and melting curve analysis. Melting curve analysis is needed to differentiate between the dermatophyte species. The species of *Trichophyton* that can be identified with the kit are *T. mentagrophytes*/*T. interdigitale* including *T. indotineae* (recently proposed), *T. mentagrophytes* (type IV), *T. rubrum*/*T. soudanense*, *T. tonsurans*, *T. violaceum*, and *T. quinckeanum*/*T. schoenleinii*. A melting peak at specific temperatures (T_m-value) allows species identification with the DermaGenius® Resistance PCR assay. Positive and negative template controls were included in each PCR run. Data analysis was performed using the Mic qPCR software (v 2.10.0), and automatic Ct analysis/T_m-range calculations were enabled. Data interpretation was done automatically using the parameter 'Genotype' in the Mic qPCR software, which resulted in dermatophyte species identification with a specific predefined T_m-values for *T. mentagrophytes/interdigitale* (57.0-60°C), *T. mentagrophytes* type IV (60-63°C), *T. tonsurans* (63.5-66.5°C), *T. quinckeanum*/*T. schoenleinii* (54-57°C) in the yellow channel and *T. rubrum/soudanense* (57.5-60.5°C), *T. violaceum* (67.5-70.5°C) in the orange channel.

2.3 Detection of terbinafine resistance as SQLE wild type and non-wild type: DermaGenius® Resistance real-time PCR kit assay, using melting-curve analyses was performed to differentiate wild-type (terbinafine susceptible) from SQLE mutant *Trichophyton* strains harbouring Phe397Leu or Leu393Phe amino acid substitutions. Each isolate was tested in duplicate and a template control (blank) was included in each run to exclude contamination. An isolate was considered positive when one of the duplicates showed increased fluorescence above the threshold. The positive control from the assay was used as a standard for the melting peaks and was tested simultaneously with the isolates to determine whether the melting peak represents wild-type or *Trichophyton* species with resistance associated mutations. Automatic data interpretation was done using the parameter ‘Genotype’ in the Mic qPCR software resulting in specific T_m-values for wild-type (66-69°C) or Resistant (55-66°C) in the Green channel.

No ethical clearance was required as the study is isolate based and no patient data is included.

3. RESULTS

The results of DermaGenius® Resistance real-time PCR assay, antifungal susceptibility testing and *SQLE* gene sequencing are summarized in Table 1. The newly developed DermaGenius® Resistance real-time PCR assay yielded following species identification; *T. mentagrophytes/interdigitale* (n=90), *T. rubrum/soudanense* (n=3), *T. mentagrophytes* (n=2), and *T. tonsurans* (n=2). The results were in concordance with those obtained from identification using ITS rDNA BLAST analysis. Importantly, in GenBank, NCBI BLAST results of the ITS region sequencing of *T. mentagrophytes/interdigitale* isolates yielded 99.5% score identity with *T. indotineae*, a newly proposed name for Indian *T. mentagrophytes/interdigitale* isolates causing recalcitrant dermatophytosis. Further, out of 90 *T. indotineae* isolates, DermaGenius® Resistance real-time PCR assay accurately identified all 40 (44%) isolates with mutant genotype. Melting curve analysis is depicted in Fig 1. Also, the newly developed real-time PCR assay detected one mutant genotype among three *T. rubrum* isolates. Overall, SQLE mutant genotypes were detected in 42.2% (n=41) isolates. Importantly, SQLE mutant genotypes obtained by DermaGenius® Resistance real-time PCR assay corroborated with *SQLE* gene sequence analysis. Of the 40 SQLE mutant *T. indotineae* isolates detected by DermaGenius® Resistance real-time PCR assay, 85% (n=34) exhibited amino acid substitution Phe397Leu and 15% (n=6) had Leu393Phe by *SQLE*

gene sequencing. All of the 40 mutant *T. indotineae* isolates had high terbinafine MICs ranging from 16- $\geq$ 32 mg/L.

4. DISCUSSION

The newly developed DermaGenius® Resistance real-time PCR assay for detection of terbinafine resistant *Trichophyton* isolates, evaluated in the present study, showed excellent correlation with *in vitro* high terbinafine MICs and *SQLE* gene sequencing mutation analysis. This data suggests that the assay is highly promising in detecting terbinafine resistance in *Trichophyton* isolates. DermaGenius® Resistance real-time PCR is a rapid assay based on melting curve analysis for species level identification and differentiation of *SQLE* wild-type from mutant genotype. Considering the worldwide surge in difficult-to-treat recalcitrant dermatophytosis due to terbinafine-resistant *Trichophyton* species, there is a paradigm shift in the treatment and diagnosis of superficial fungal infections.^{3,19} The key to effective antifungal therapy for a recalcitrant infection is isolation and identification of the dermatophyte followed by susceptibility testing of the strain against antifungal drugs. However, it is important to emphasize that susceptibility testing of dermatophytes is not a routine practice worldwide, so terbinafine resistance is probably underestimated.^{4, 19} Accurate molecular identification and antifungal susceptibility profiling of the strain increase the turnover time for treatment of dermatophytosis. Further, antifungal susceptibility testing of dermatophytes requires prolonged incubation (up to 7 days), has been difficult to standardise due to technical difficulties, and has suboptimal reproducibility. However, recently, European Committee on Antimicrobial Susceptibility Testing (EUCAST) has established a standardised microdilution method (E.Def 11.0) for microconidia forming dermatophytes. The method recommends using chloramphenicol and cycloheximide supplemented growth media to circumvent bacterial and saprophytic mould contamination and spectrophotometric endpoint reading using 50% growth inhibition. The method will help in generating reproducible *in vitro* susceptibility data of dermatophytes in future studies.^{33, 34}

Novel rapid diagnostics targeting detection of antifungal resistance are warranted to overcome the rising epidemic of antifungal resistance. The newly developed DermaGenius® Resistance real-time PCR assay is highly efficacious and demonstrated both high-throughput identification of *Trichophyton* species and differentiation of *SQLE* wild-type (susceptible) from mutant genotypes.

Thus, the assay decreases the turnover time as both identification and determination of resistance could be targeted in 3 hrs, thereby resulting in early selection of correct treatment options. However, it is essential to address that this assay does not differentiate between the substituted amino acid occurring in the *SQLE* mutant *Trichophyton* isolates. Thus, the assay has limited application in generating geographic molecular epidemiologic data warranted to estimate the prevalence of specific mutations in the terbinafine target *SQLE* gene. In this study, isolates that harboured Phe397Leu or Leu393Phe were analysed for detection of *SQLE* mutants which were accurately detected as non-wild type by DermaGenius® Resistance real-time PCR assay, however, the assay did not differentiate between the two mutations. Furthermore, extension of the DermaGenius® Resistance real-time PCR assay for detection of other reported mutations is warranted. Also, it is important to consider implementing this assay for direct detection of *SQLE* mutants in clinical samples, without previous culture, which will provide significant reduction of the turnover time to effectively treat patients with recalcitrant dermatophytosis.³⁵ The culture independent approach to detect resistant mutations directly in clinical specimens has been previously reported in aspergillosis patients. AsperGenius® Resistance multiplex real-time PCR assay to detect azole resistant mutations in *Aspergillus fumigatus* directly in bronchoalveolar lavage (BAL) samples of 160 patients with chronic respiratory diseases in Delhi, India showed promising results. In culture-negative, PCR-positive samples, azole-resistant mutations were detected in 34% of BAL samples suggesting high sensitivity of the assay to detect resistance associated mutations.³⁵ Finally, in the present study the DermaGenius® Resistance real-time PCR assay proved to be an effective and robust method, however, further studies to assess its efficacy with other *Trichophyton* species are warranted.

5. DATA AVAILABILITY STATEMENT

The ITS and *SQLE* gene sequences used in the present study were previously published and are accessible at GenBank (<https://www.ncbi.nlm.nih.gov/genbank/>) through accession numbers MH114933 to MH114952 and MH990799-MH990859.

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Table 1: Genotypic characteristics of *Trichophyton* species isolates (n=97) subjected to molecular identification using ITS rDNA sequencing, SQLE amino acid substitution and DermaGenius® Resistance real-time PCR assay.

Diagnostic and mutation parameters	<i>Trichophyton</i> species (n=97)			
DG [†] Identification	<i>T. interdigitale/mentagrophytes</i> (n=90)	<i>T. rubrum/soudanense</i> (n=3)	<i>T. mentagrophytes</i> (type IV) (n=2)	<i>T. tonsurans</i> (n=2)
ITS sequencing Identification	<i>T. indotineae</i> [‡] (n=90)	<i>T. rubrum/soudanense</i> (n=3)	<i>T. mentagrophytes</i> (n=2)	<i>T. tonsurans</i> (n=2)
DG SQLE [§] Analysis				
Wild type/mutant	WT [¶] (n=50)/mutant (n=40)	WT (n=2)/mutant (n=1)	WT (n=2)	WT (n=2)
Amino acid substitutions in SQLE ^p by sequencing				
1. Wild type	WT (n=50)	WT (n=2)	WT (n=2)	WT (n=2)
2. Mutant	(Phe397Leu, n=34; Leu393Phe, n=6)	(Phe397Leu, n=1)	-	-

[†]DermaGenius® Resistance real-time PCR assay.

[‡]BLAST analysis of *T.interdigitale/mentagrophytes* isolates using GenBank showed 99.5 % identity with *T. indotineae* ITS rDNA sequences submitted by Kano *et al*, 2020.

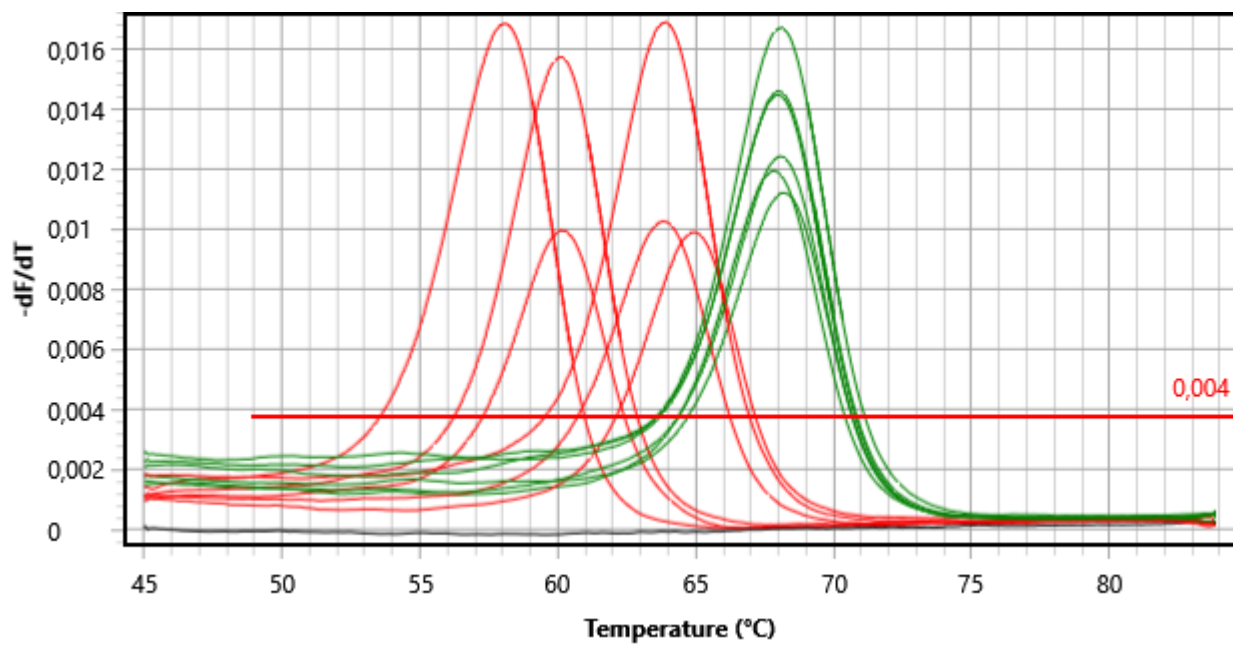
[§]SQLE, squalene epoxidase

[¶]WT: wild type

Terbinafine MIC (minimum inhibitory concentration) Range: 0.06-1 mg/L recorded in Wild Type; 16->32 mg/L recorded in SQLE Mutants

Figure Legend

Figure 1: Graphical representation of melting curve analysis depicting change in fluorescence versus temperature (°C). The peaks depict differentiation between wild type (green peaks) and terbinafine target *SQLE* gene mutant genotype (red peaks).



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